STEP generation · timing and counts

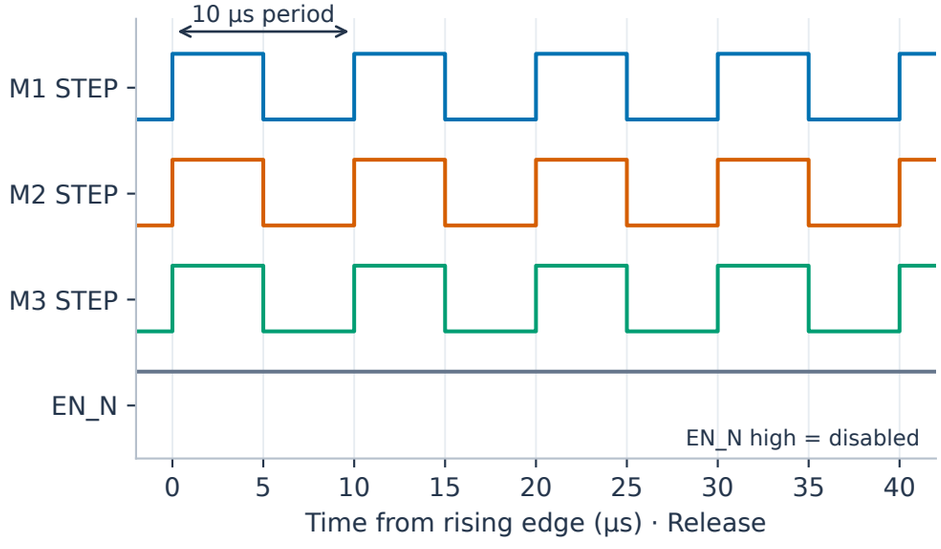
ST-Link crystal → 5 MHz HSE bypass | CPU 80 MHz | TIM2 10 MHz | Analyzer 24 MHz

18 / 18 checks passed

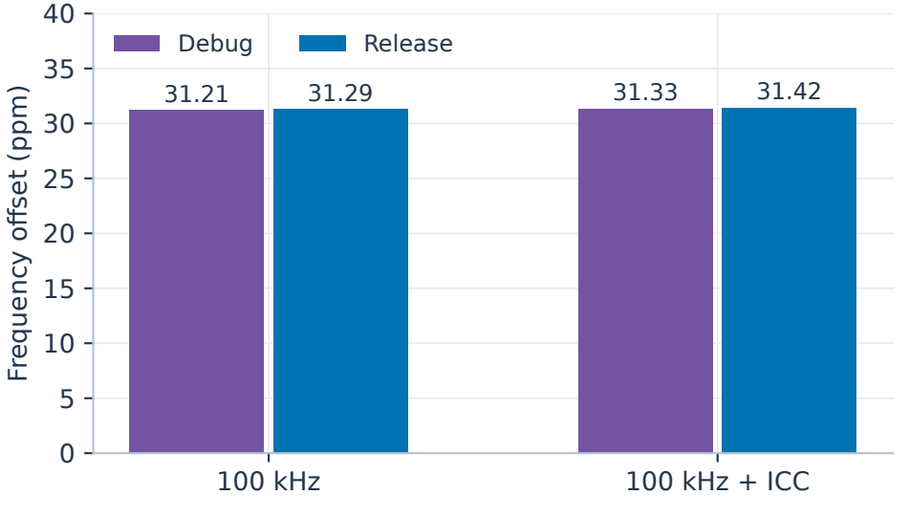
	Debug	Release
1 · Exact 1 / 2 / 3 pulses	PASS	PASS
2 · Independent starts	PASS	PASS
3 · 3 × 100 kHz	PASS	PASS
4 · Rate update	PASS	PASS
5 · M2 stop / restart	PASS	PASS
6 · Timer rollover	PASS	PASS
7 · Underrun guard	PASS	PASS
8 · 3 × 100 kHz + ICC	PASS	PASS
9 · 1,000-pulse profile	PASS	PASS

1,290,235 captured pulses · counts match

100 kHz on all three axes · ICC active

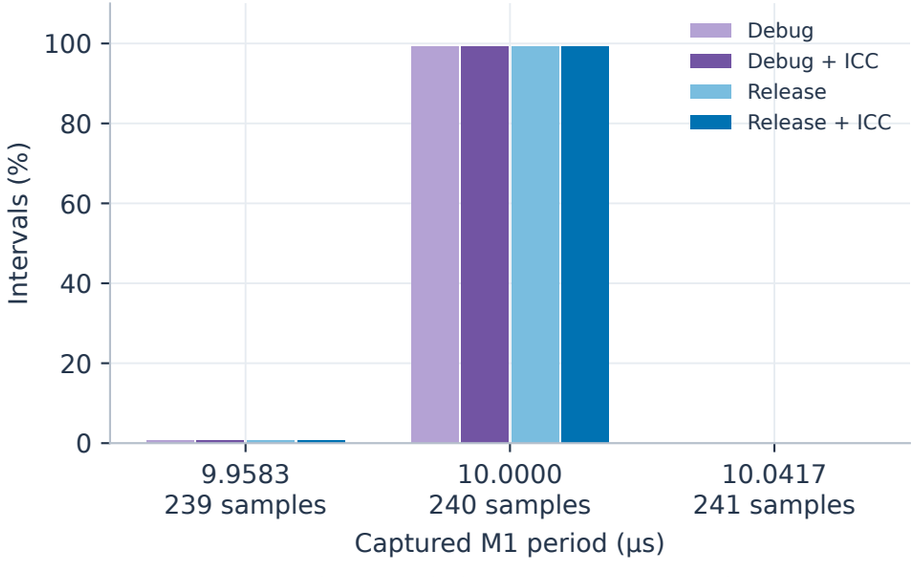


Measured frequency · M1–M3 equal



≈ +0.0031% relative to the uncalibrated analyzer

Period distribution · 99,999 intervals / run

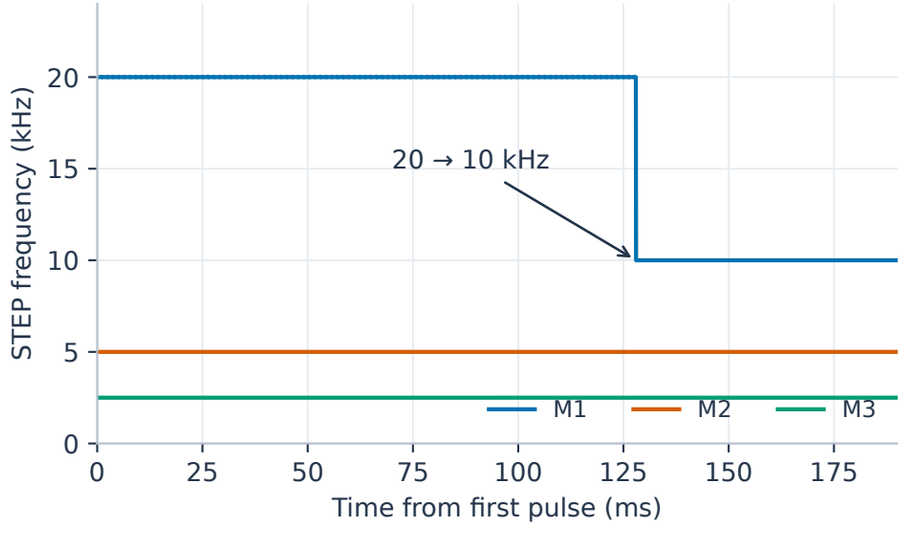


Analyzer resolution: 41.67 ns per sample

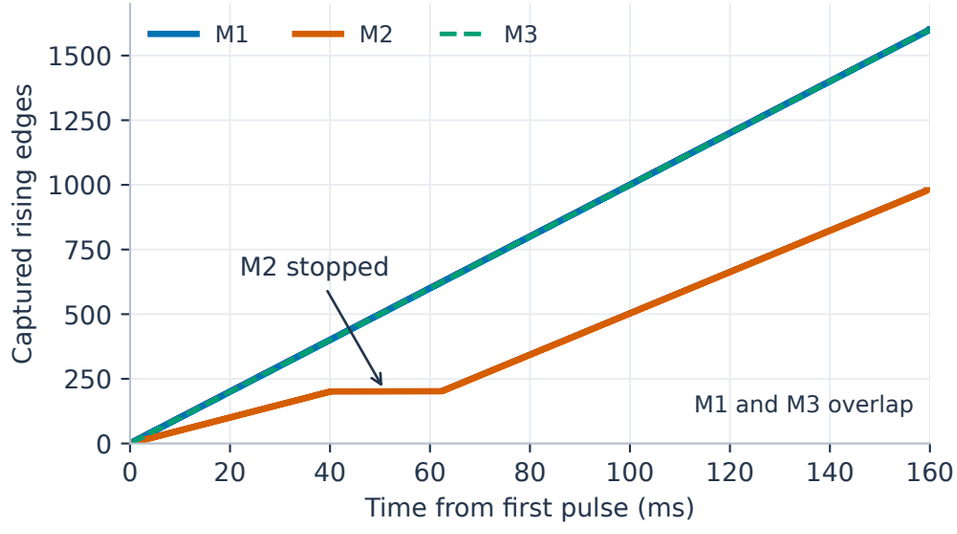
STEP generation · motion and reset

Release captures | Nominal STEP high time 5 µs | EN_N remains high

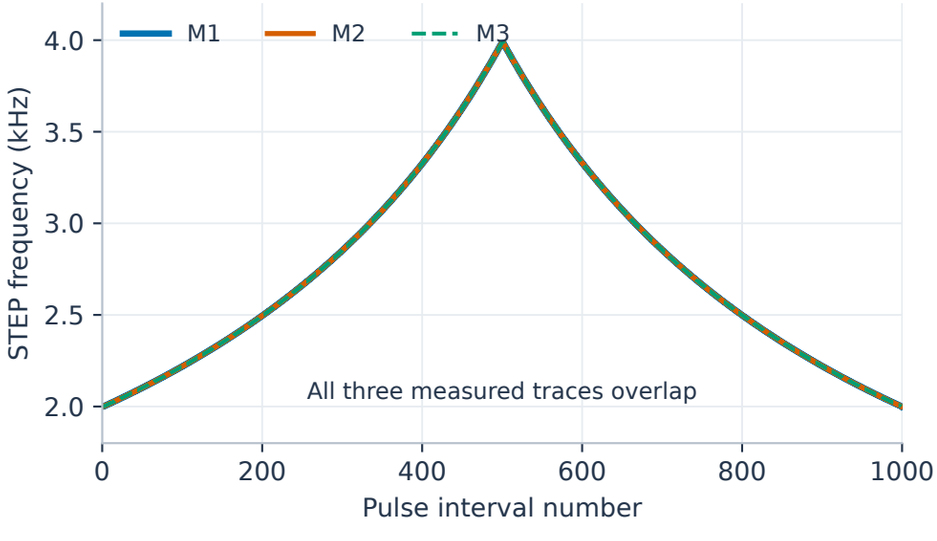
M1 rate update · other axes continue



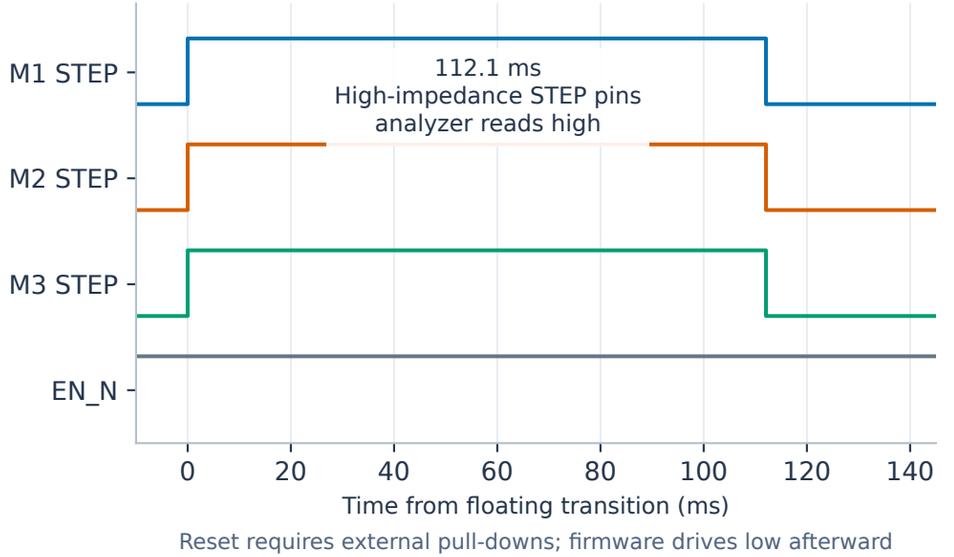
Stop / restart M2 independently



Varying profile · 1,000 pulses per axis



Reset · bare Nucleo pins



Reset requires external pull-downs; firmware drives low afterward